

Quick Revision Worksheet

Topic: Capacitors & RC Circuits

Target Level: Grade 12 (GS + LS)

Instructions: Choose the best answer. For calculation questions, clearly show your work in the space provided.

1. An initially uncharged capacitor is connected in series with a resistor and an ideal battery. **At the exact moment the switch is closed ($t = 0$), what is the behavior of the capacitor?**
 - A. It acts as an open circuit ($i = 0$).
 - B. It acts as a pure resistor with resistance R .
 - C. It acts as a short circuit ($u_C = 0$).
 - D. It immediately reaches the battery voltage E .
2. A capacitor with $C = 220 \mu\text{F}$ is connected to a 10 V DC source. **What is the maximum electrical energy stored in the capacitor at steady state? (Show your work)**
 - A. 11.0 mJ
 - B. 22.0 mJ
 - C. 1.1 J
 - D. 110 μJ
3. In an RC charging circuit, the differential equation for the capacitor voltage is given by $u_C + \tau \frac{du_C}{dt} = E$. **What does the term $\tau \frac{du_C}{dt}$ represent physically when $t = 0$?**
 - A. The steady-state voltage across the capacitor.
 - B. The initial voltage across the resistor u_R .
 - C. The total energy stored in the dielectric.
 - D. The internal resistance of the battery.
4. According to the "Golden Rule of Symmetry", **the capacitor resists a change in voltage u_C exactly like the coil (inductor) resists a change in:**
 - A. Current (i)
 - B. Resistance (R)
 - C. Magnetic Flux (Φ)
 - D. Electromotive force (E)
5. A student is determining the time constant τ graphically. **If the tangent to the charging curve at $t = 0$ is drawn, where will it intersect the horizontal asymptote $u_C = E$?**
 - A. At the origin $(0, 0)$.
 - B. At the voltage $u_C = 0.63E$.
 - C. At the time $t = 5\tau$.
 - D. At the time $t = \tau$.

6. During the discharge of a capacitor through a resistor R , the voltage follows $u_C(t) = Ee^{-t/\tau}$. **If we linearize this by plotting $\ln(u_C)$ versus time t , what is the slope of the resulting line?**
- $m = \tau$
 - $m = -1/\tau$
 - $m = -RC$
 - $m = \ln(E)$
7. In a charging RC circuit, **at what time t do the voltage across the capacitor (u_C) and the voltage across the resistor (u_R) have exactly the same value?** (*Show your work*)
- $t = \tau \ln(2)$
 - $t = \tau$
 - $t = 0.5\tau$
 - $t = \tau \ln(0.63)$
8. **What is the physical function of the dielectric material placed between the armatures of a capacitor?**
- To allow a small leakage current to prevent overcharging.
 - To act as an insulator and increase the capacity of the capacitor.
 - To decrease the time constant τ for faster charging.
 - To convert electrical energy into magnetic energy.
9. **If the resistance in a discharging loop is doubled while keeping the same capacitor, how does the discharge time change?**
- The discharge time remains unchanged.
 - The discharge occurs twice as fast (τ is halved).
 - The capacitor will never fully discharge.
 - The discharge takes twice as long (τ doubles).
10. During the complete discharge phase of an RC circuit, **what happens to the energy $W_e = \frac{1}{2}Cu_C^2$ that was stored in the capacitor?**
- It is returned to the battery.
 - It is stored in the electric field of the dielectric permanently.
 - It is dissipated as thermal energy (Joule heat) in the resistor.
 - It is destroyed, violating the law of conservation of energy.